



Canon Pixma iP 1000

The HP 3845 lost out to HP 3745 mainly due to its high price-per-page cost. The Best Buy Gold award goes to Canon Pixma iP1000 and the Best Buy Silver goes to HP DeskJet 3745.



HP DeskJet 3745



Office And Photo Printers

In this category, we had eight printers: the Canon Pixma iP1500, iP2000, iP3000, and iP4000; Epson Photo R210 and R310; the HP DeskJet 5748; and Lexmark P915, all priced over Rs 4,000.

Features

Features carried weightage, as did performance while, price was given slightly lower weightage.

The features we got to see on some printers really brought smiles to our faces. Soon enough, it became clear that it would be the Lexmark that would set the benchmark in terms of its sheer feature list, ease of use and looks.

When we unpacked the P915, we heard 'oohs' and 'aahs', with people flocking to the desk all the time trying to play with its colour screen. The Lexmark probably kept ease of use as the No. 1 priority while designing this printer.

The setting up was hassle-free, and there was even an animated display showing us how to replace the cartridges. The ink status display had a 'skinnable' interface and showed the display in real time. It supports PictBridge, and has slots for various memory cards. It even lets you do simple photo editing:

How We Tested

Our test bed comprised a Pentium 4 3.2 GHz processor plugged onto an MSI 875P Neo FISR2 motherboard with 512 MB of Corsair 400 MHz DDR RAM, a Gainward nVidia GeForce FX 5950 graphics card, and an 80 GB 7200 rpm ATA IV Seagate Barracuda hard disk.

We used Windows XP with SP1 as the OS. The system was loaded with all the latest chipset and graphics drivers. We also installed the USB patch for XP and connected the devices on the USB 2.0 port for optimum performance. Across the tests, we used the following media:

Text/combi-document printouts:

100 g/sm paper from Berga (A4 size)

Photo Printouts: 270 g/sm glossy photo paper from Kodak (A4 size)

Inkjet Printers

We divided printers into two categories: Home printers (sub-Rs 4,000) and Office/Photo printers (Rs 4,000 and above). The test process remained same for both groups; the only change was in the parameters to determine the winner.

Features

Printers were rated on various features: PictBridge, memory card readers, duplex printing, CD/DVD printing support, input/output tray capacity, input buffer and maximum resolution supported by the printers' LCD screens. They were rated on the basis of features they supported. Obviously, most of these features were standard on the higher-end printers.

Performance

We tested the performance on three parameters; speed, quality, and efficiency.

Economy Test

We conducted the economy test first to avoid miscalculations later, with half-full cartridges. We created a document that had eight per cent of text and 60 per cent of images and graphs. We then printed multiple copies of the same drain test document on regular paper. We noted an average ten per cent drop in ink and extrapolated it to get an estimate of the printer's ink usage.

Speed And Quality Test

We used a text document to test the raw print speeds of the printer. The quality settings used were Normal and Best.

To test the ability of printers to handle various aspects of a regular document, we created a combi-document with black text, interspersed with graphs and images. We clocked the time it took them to print this document in normal and best quality modes, and accordingly analysed the print quality.

The final test for printers was the photo test, where we used 270 g/sm Kodak Glossy Photo Paper to print an A4 image. We timed the process and examined it for quality of print. The settings used here were the highest quality available.

Inkjet MFDs

MFDs have a scanner, printer and copier. MFDs with fax facility earned more points.

Printer Test

The test process here was exactly the same as that for the inkjet printers.

Scanner Test

We tested the scanners for speed using an average of five previews of an A4 size image. This was done when the scanner was fresh

out of the box to incorporate the warm-up time. Scanners with a higher warm-up time didn't fare as well as the others.

We tested the warmed up scanners to negate any possibility that a scanner might scan faster than the others once warmed up. We imported the same A4 size image at 150 dpi and 600 dpi, in Adobe Photoshop, and scanned a full text document in B&W mode at 200 dpi. The scanner then underwent a test to determine its colour differentiating abilities using a Kodak IT8 card with different shades of colour—each with a little difference from the preceding one.

Next, we tested the resolution card to figure actual scanning capabilities by checking for its native resolutions. We scanned the card at 300 dpi to see how many boxes it could differentiate in between the lines. Most MFDs passed this test easily.

Finally, we did an OCR test to check the scanner's ability to differentiate light and dark areas. A print article was scanned into the trial version of Abbyy Fine Reader Pro 7.0, and scanned for wrongly-interpreted letters in the software. We then gave a percentile value to the devices.

Copier

We copied a text document in B&W and the combi-document in colour, and rated the MFDs on their speed and quality of printouts.

How The Awards Are Given

The scores from features, performance and price are given category-relevant weightage. An overall score out of 100 is calculated. The product that scores the highest here is adjudged the winner of the Best Buy Gold award for this category. The second highest score gets the Digit Best Buy Silver award.